TokoBli Campaign Evaluation & Product Page Analysis

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Business Overview & Disclaimer

Company Overview

TokoBli is Indonesia's leading e-commerce platform, offering diverse product categories and running major sales campaigns on “twin dates”. The business aims to leverage data analytics to optimize campaign strategies, grow transactions, and maximize revenue while keeping campaign costs efficient.

Disclaimer

Analysis performed on provided e-commerce dataset for educational demonstration. Insights and recommendations are based strictly on TokoBli dataset.

Project Goal

1. Evaluate the previous three campaign scenarios and provide actionable insights.
2. Experiment whether the new product detail page will bring more transactions than the current product detail page.

Dataset Overview

Main tables and essential fields

Methodology

Approach

- Data understanding, cleaning (removal of duplicates / missing / inconsistent types / outliers), descriptive analytics
- Exploratory Data Analysis (EDA) to compare all main metrics per campaign
- Statistical hypothesis testing to assess significance of differences in product detail page

Tools Used

Google Sheets (with visualizations, basic statistical formulas), descriptive/EDA charts, hypothesis testing models.

Link spreadsheet : https://docs.google.com/spreadsheets/d/1MAo2gzINaGUGjHymLv326WHolFJUFyIc/edit?gid=1238598707#gid=1238598707

Key Steps

Lay out your workflow clearly (clearly state each key business question to be addressed).

1. Business Understanding

Question

A. Define a clear, measurable problem statement for TokoBli, considering the company's business operations and specific limitations (using the SMART criteria).

Answer

The business questions is: **"Which major sales campaign on twin date – 10/10, 11/11, or 12/12- yields the best business results based on transaction data from past three campaigns period ?"**

The e-commerce platform runs major sales campaigns on "twin dates" such as 10/10, 11/11, and 12/12, each campaign has a distinct scenario.

Thus, we want to determine which scenario yields the best business results.

S	Specific – target a specific area for improvement	Business results from from past three campaigns 10/10, 11/11, & 12/12
M	Measurable – quantify or show an indicator of progress Applied to the ideal state:	Measurable using metrics 1. Campaign budget for discounts 2. Transaction count 3. Customer count 4. Product sold 5. Total revenue
A	Achievable – they need to be agreed, to be attainable and able to be implemented	Result is already achieved
R	Realistic – states what results can realistically be achieved, given available resources	Realistic from past data
T	Time-bound - there need to be deadlines, but are they reasonable?	Past three campaigns period

1. Business Understanding

Question

B. Identify and select the most relevant business metrics from the available data that best address the problem statement (e.g., average revenue per campaign, total transactions, etc.).

Answer

This metric shows how much **cost** for the platform to run the sales campaign:

1. Campaign budget for discounts

These metrics show how much gain the platform get from running the sales campaign:

2. Transaction count per campaign : gain in terms of number of transaction during campaign period
3. Customer count per campaign : gain in terms of number of customer during campaign period
4. Product sold per campaign : gain in terms of number of product sold during campaign period
5. Total revenue per campaign : gain in terms of number of revenue during campaign period

2. Clean Up The Data

Question

A. Removing or imputing missing values in rows and columns (delete columns if over 50% missing, otherwise handle rows).

Answer

Column:

Missing values >50% in a column are present in “Status” column

1. =COUNTBLANK(A2:A10160)/10159
2. Change data type to percentage
3. Drag from A to N
4. **Delete “Status” column**

A	B	C	D	E	F	G	H	I	J	K	L	M
Transactior	Product ID	Customer ID	Status	Created At	Campaign	SKU	Price	QTY	Discount	Shipping Cos	Total Revenue	Category Name
ID-13046	830530	104877		2023-12-12	Campaign 12/12	MEFURC5AD9BB4312C23	9,00E+05	3	0,00E+00	0	2.700.000	Men's Fashion
ID-13047	830413	105097		2023-12-12	Campaign 12/12	MEFMYD5A9E5E30E80F7	6,49E+06	1	0,00E+00	0	6.490.000	Men's Fashion
0%	0%	0%	99%	0%	0%	0%	0%	0%	0%	0%	0%	0

Missing Values

Transaction ID	Price	QTY	Discount
ID-147		1	0,00E+00
ID-152	1,76E+06		0,00E+00
ID-177	1,99E+06	1	

After Impute

Transaction ID	Price	QTY	Discount
ID-147	2,50E+06	1	0,00E+00
ID-152	1,76E+06	1	0,00E+00
ID-177	1,99E+06	1	1,99E+05

LANJUTAN

Row:

1. Tambahkan kolom bantu untuk hitung jumlah sel kosong per baris
2. =COUNTBLANK(A2:M2) di sel O2
3. Pilih sel O2.
4. Tekan Shift + klik 010160 untuk seleksi sampai baris terakhir
5. Tekan Ctrl + D untuk “Fill Down” rumus ke semua baris.
6. Filter

Missing values are present in ID-147, ID-152, ID-177.

We can **impute** the missing values using simple business logic.

Price =(Total Revenue + Discount - Shipping Cost) / Qty

Qty =(Total Revenue + Discount - Shipping Cost) / Price

Discount=Price * Qty - Total Revenue + Shipping Cost

2. Clean Up The Data

Question

B. Eliminating duplicate rows.

Answer

Duplicated data :
Transaction ID (ID-164, ID-269, ID-303, ID-546, ID-12769).
Those UserID should be eliminated.

Transaction ID	Product ID	Customer ID	Created At	Campaign	SKU	Price	QTY	Discount
ID-164	725812	15124	2023-10-10	Campaign 10/10	KABDYN5A0029BC637C1	1,99E+06	1	0,00E+00
ID-164	725812	15124	2023-10-10	Campaign 10/10	KABDYN5A0029BC637C1	1,99E+06	1	0,00E+00
ID-269	726147	48507	2023-10-10	Campaign 10/10	BAGDEA5A23FADEEAA56	6,99E+06	1	0,00E+00
ID-269	726147	48507	2023-10-10	Campaign 10/10	BAGDEA5A23FADEEAA56	6,99E+06	1	0,00E+00
ID-303	726117	84712	2023-10-10	Campaign 10/10	MEFWIN5A263D6DC789C	5,49E+06	1	0,00E+00
ID-303	726117	84712	2023-10-10	Campaign 10/10	MEFWIN5A263D6DC789C	5,49E+06	1	0,00E+00
ID-546	727553	84831	2023-10-10	Campaign 10/10	MEFBNB5A046DA417C54-M	4,49E+06	1	0,00E+00
ID-546	727553	84831	2023-10-10	Campaign 10/10	MEFBNB5A046DA417C54-M	4,49E+06	1	0,00E+00
ID-12769	829033	105375	2023-12-12	Campaign 12/12	MEFMYD5A9E5E30E80F7	6,49E+06	1	0,00E+00
ID-12769	829033	105375	2023-12-12	Campaign 12/12	MEFMYD5A9E5E30E80F7	6,49E+06	1	0,00E+00

2. Clean Up The Data

Question

C. Correcting data types

Answer

Columns with wrong data type are “Price”, “Discount(%)”

Price	Discount
2,10E+06	0,00E+00
4,20E+06	0,00E+00
1,20E+06	0,00E+00

Price	Discount
2100000	0
4200000	0
1200000	0

3. Descriptive Analytics

Question

A. Identify and remove outliers based on the "Total Revenue" column using appropriate statistical methods (apply across all campaigns, not by each campaign).

Answer

Total Revenue	Q1	Q3	IQR	Upper Outlier	Lower Outlier
	2241000	6987500	4746500	14107250	-4878750

- Lower outlier negative value → No lower outlier
- Delete rows with Total Revenue as in this table
- In the end, we will have 10143 rows of clean data.

Total Revenue
78.400.000
69.900.000
64.000.000
50.190.000
48.090.000
16.590.000
74.250.000
74.250.000
21.000.000
83.556.000
65.100.000
69.800.000

3. Descriptive Analytics

Question

B. Calculate statistical measurements (count, minimum, maximum, mean, median, mode, range, Q1, Q3, variance, standard deviation, skewness) for Quantity (QTY), Discount, and Total Revenue.

Answer

QTY	Campaign 10/10	Campaign 11/11	Campaign 12/12
Mean	1	1	1
Median	1	1	1
Mode	1	1	1
Standard Deviation	1	1	1
Sample Variance	2	1	1
Skewness	18	22	3
Range	29	29	4
Minimum	1	1	1
Maximum	30	30	5
Count	3329	3399	3414
Q1	1	1	1
Q3	1	1	1

- All campaigns have a median, mode, Q1, and Q3 = 1
- This means **most transactions sold only 1 piece per transaction, across all campaigns.**
- The distribution is highly concentrated at 1, suggesting either:
  Many low-volume SKUs dominate sales
  OR
  Customers tend to buy single units per item

DISCOUNT

Discount	Campaign 10/10	Campaign 11/11	Campaign 12/12
Mean	82886	89037	74058
Median	0	0	0
Mode	0	0	0
Standard Deviation	184135	197027	179486
Sample Variance	33905771990	38819497477	32215129511
Skewness	2	2	3
Range	900000	899000	899000
Minimum	0	0	0
Maximum	900000	899000	899000
Count	3329	3399	3414
Q1	0	0	0
Q3	20000	40000	0

- **Most products received no discount**
- Median, Mode, Q1 = 0 across all campaigns → majority of products had no discount at all
- **Campaign 11/11 received highest discount among other campaign**

TOTAL REVENUE

Total Revenue	Campaign 10/10	Campaign 11/11	Campaign 12/12
Mean	4461986	4890302	4335479
Median	3990000	5200000	3990000
Mode	6990000	7200000	2990000
Standard Deviation	2644626	2637724	2701886
Sample Variance	6994044361764	6957590470647	7300189181921
Skewness	0	0	1
Range	13620000	13586000	13895000
Minimum	180000	189000	45000
Maximum	13800000	13775000	13940000
Count	3329	3399	3414
Q1	2320000	2490000	1990000
Q3	6837750	7200000	6300000

- **Campaign 11/11 generated the highest average and median revenue**
- 11/11 outperforms in both central tendency and upper quartile indicates stronger per-transaction revenue, possibly due to higher-value SKUs or better bundling
- All campaigns show high variance and standard deviation
- Revenue per transaction is highly variable, which is typical in e-commerce

3. Descriptive Analytics

Question

C. Provide a minimum of 3 key business insights based on the statistical analysis results.

Answer

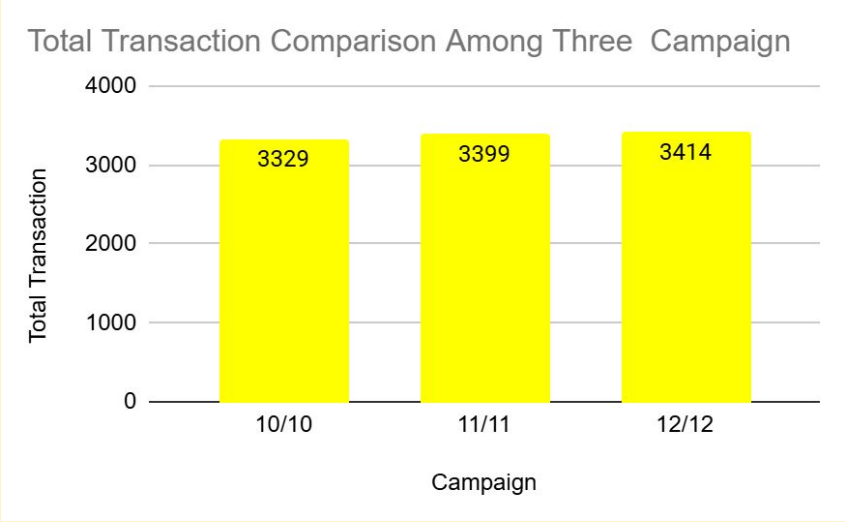
1. Most transactions sold only 1 piece per transaction, across all campaigns.
2. Most products received no discount
3. Campaign 11/11 generated the highest average and median revenue but also received highest discount among other campaign

4. Exploratory Data Analysis

Question

A. Determine whether there are differences in total transactions, total customers, total products sold, total revenue, and total discount campaign budget among the three campaign periods. Present summary tables and charts, then discuss insights.

Answer

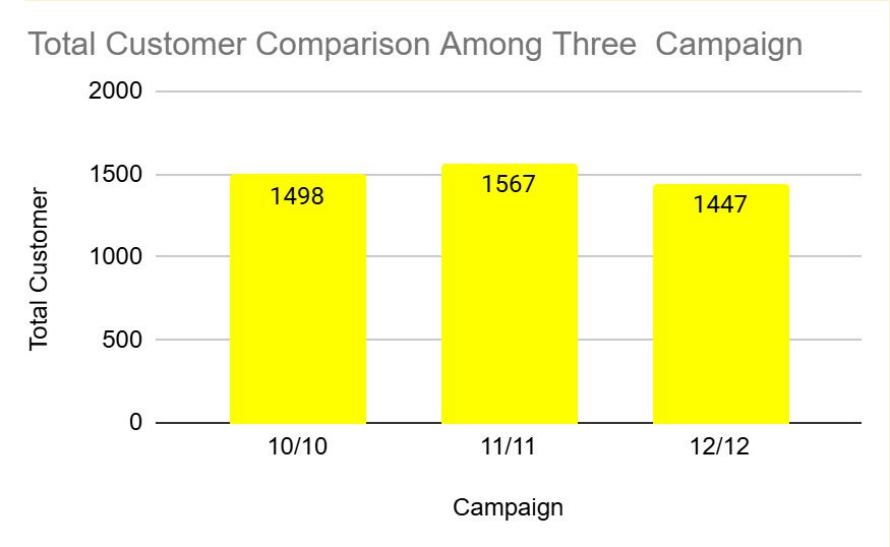


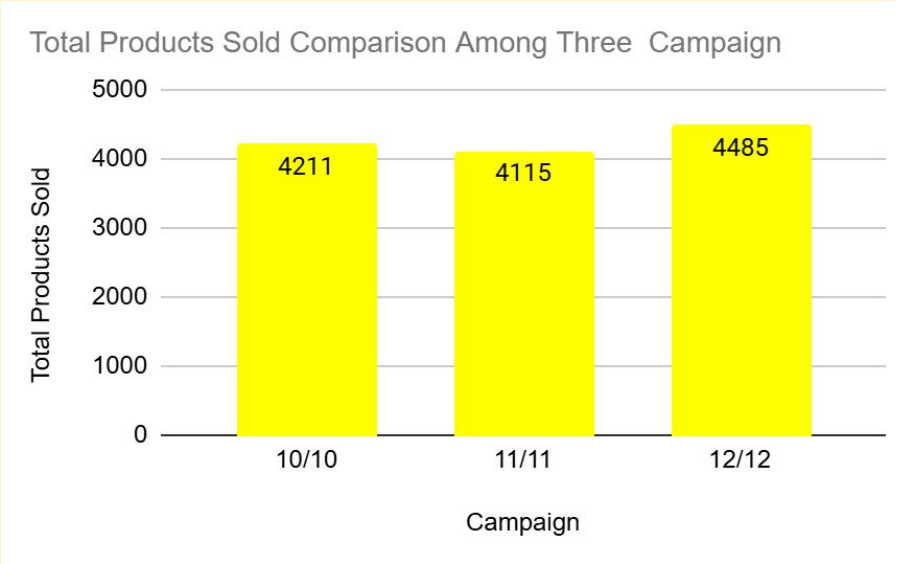
Campaign	10/10	11/11	12/12
Total Transaction	3329	3399	3414

- Campaign 12/12 has the highest total transaction among other campaign

4. Exploratory Data Analysis

Answer





Campaign	10/10	11/11	12/12
Total Customer	1498	1567	1447

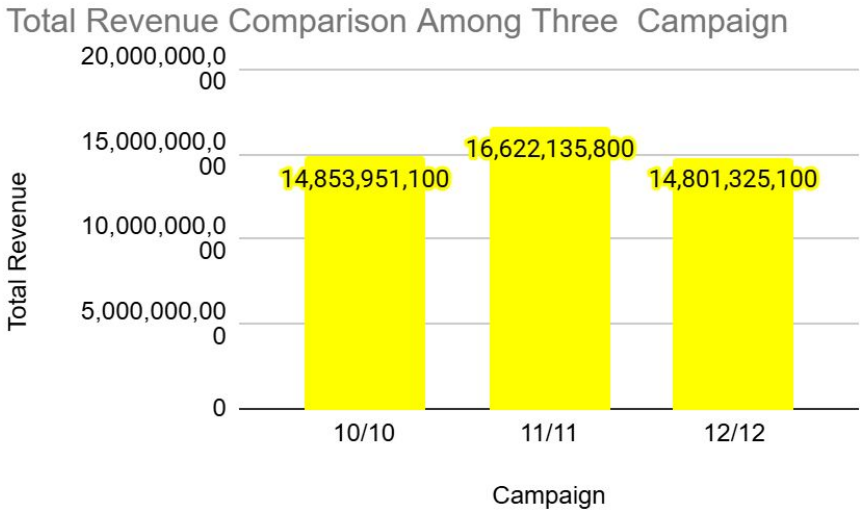
- Campaign 11/11 has the highest Total Customer among other campaign

Campaign	10/10	11/11	12/12
Total Products Sold	4211	4115	4485

- Campaign 12/12 has the highest Total Products Sold among other campaign

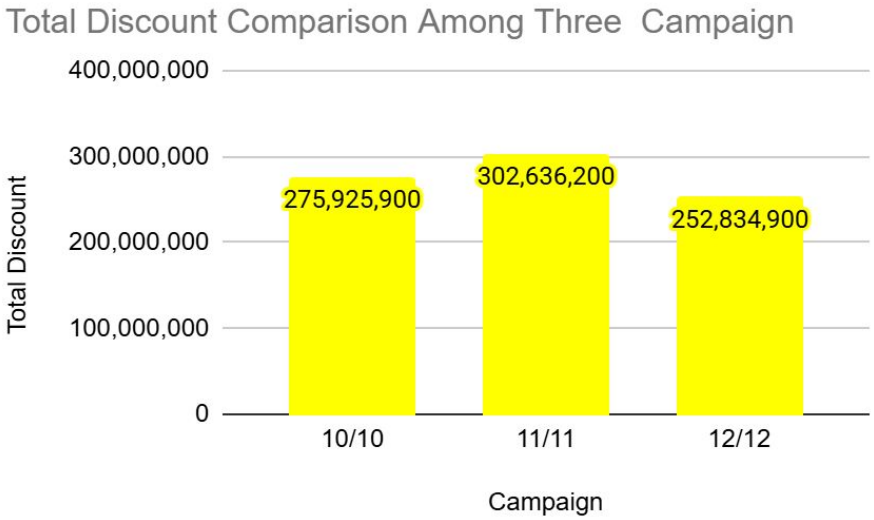
4. Exploratory Data Analysis

Answer



Campaign	10/10	11/11	12/12
Total Revenue	14,853,951,100	16,622,135,800	14,801,325,100

- Campaign 11/11 has the highest Total Revenue among other campaign



Campaign	10/10	11/11	12/12
Total Discount	275,925,900	302,636,200	252,834,900

- Campaign 11/11 has the highest Total Discount Campaign Budget among other campaign

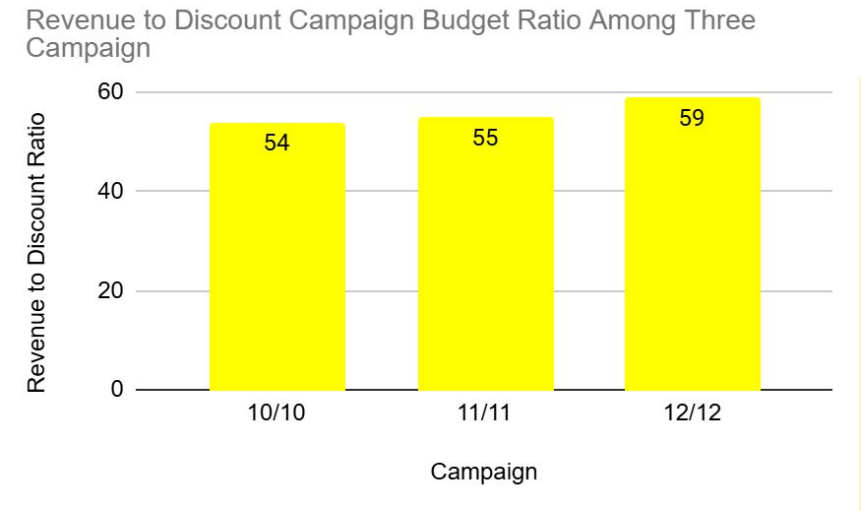
4. Exploratory Data Analysis

Question

B. Evaluate differences in the ratio of revenue to discount campaign budget for each campaign period, show calculations, charts, and insights.

Answer

- Campaign 12/12 has the highest Revenue to Discount Campaign Budget Ratio among other campaign



Campaign	10/10	11/11	12/12
Total Revenue	14,853,951,100	16,622,135,800	14,801,325,100
Total Discount	275,925,900	302,636,200	252,834,900
Revenue to Discount Ratio (Total Revenue / Total Discount)	54	55	59

4. Exploratory Data Analysis

Question

C. Identify which product category had the highest transactions, customers, products sold, and revenue, while also having the lowest discount campaign budget —breakdown per campaign. Present findings and business meaning.

Answer

Campaign	Highest Transactions, Customers, Products Sold, & Revenue	Lowest Discount Campaign Budget
10/10	1. Men's Fashion (highest revenue & customer) 2. Beauty & Grooming (highest transaction & product sold)	1. Others 2. Books
11/11	Men's Fashion	Books
12/12	1. Men's Fashion (highest revenue & customer) 2. Health & Sports (highest transaction & product sold)	Books

- There was no single product category had the highest transactions, customers, products sold, and revenue, while also having the lowest discount campaign budget
- Books category had lowest discount but also had the lowest revenue
- **Men's Fashion** category consistently had the **highest revenue & customer** on all campaign, while also having the **not the highest & not the lowest discount** campaign budget

4. Exploratory Data Analysis

Question

D. Find the product category with the highest ratio of revenue to discount campaign budget in each campaign period—show which category, calculations, and insights.

Answer

Campaign	Category with Highest Revenue To Discount Ratio
10/10	1. School & Education (105) 2. Men's Fashion (100)
11/11	Men's Fashion (139)
12/12	1. Books (139) 2. Men's Fashion (91)

- Calculation :
  Revenue To Discount Ratio = Total Revenue / Total Discount
  (detailed calculation on spreadsheet)

- School & Education & Books had far less revenue than Men's Fashion indicating far less sales volume

- Men's Fashion had highest ratio of revenue to discount campaign budget in each campaign period

4. Exploratory Data Analysis

Question

E. Summarize overall insights and provide actionable business recommendations based on all EDA and statistical findings.

Answer

Key Findings:

1. **Campaign 12/12** yielded the **best business results** because Campaign 12/12 had the **highest Revenue to Discount** Campaign Budget Ratio among other campaign.

2. **Men's Fashion** was the **best performing category** because Men's Fashion had **highest ratio of Revenue to Discount** campaign budget in each campaign.

3. **Most transactions** sold **only 1 piece** per transaction, across all campaigns

4. **School & Education and Books** had far **less sales volume** than Men's Fashion despite **higher ratio of Revenue to Discount** campaign budget

Insights / Observation	Business Impact	Isolation	Prioritization	Recommendation
Campaign 12/12 yielded the best business results (highest Revenue to Discount Rasio) among other campaign	Generating more revenue more efficiently	Major sales campaign scenario on twin date	High (High impact – High effort – Many data point)	Apply Campaign 12/12 scenario for future major sales campaign on twin date
Men's Fashion was the best performing category (highest ratio of Revenue to Discount) in each campaign.	Generating more revenue more efficiently	Product focus for next twin date campaign	High (High impact – High effort – Many data point)	1. Push promotion on Men's Fashion category for for next twin date campaign 2. Up sell, cross sell, & bundling strategy with complementary category, example : Beauty & Grooming category, Health & Sport category
Most transactions sold only 1 piece per transaction	Increasing revenue per transaction	Quantity per transaction	Medium (Medium impact – High effort – Many data point)	1. Bundling strategy 2. Minimum quantity per transaction for discount
School & Education and Books had low sales volume, high Revenue to Discount ratio	Boost slow moving product	School & Education and Books category	Low (Medium impact -High effort – Low data point)	1. Marketing strategy to boost sales of these categories 2. Find new market opportunities which have high demand for these categories

5. A/B Testing

Question

A. Develop Hypothesis Question for comparison between group A & group B

Answer

H_0 Null Hypothesis	There is **no significant difference** between **new** product detail page design **and current** product detail page design in transaction value per user
H_1 Alternative Hypothesis	There is **significant difference** between **new** product detail page design **and current** product detail page design in transaction value per user

- **Group A**: Users who interacted with the **current** product page design
- **Group B**: Users who were exposed to the **new** product page design

5. A/B Testing

Question

B. Provide Test Chosen (e.g., t-test) & Assumptions

Answer

Chosen test : t-Test: Two Sample Assuming Equal Variances

- t-Test because the population variance is unknown
- Equal variances because Variance Ratio ≤ 4

Variance Group A	Rp22.436.486.773,86
Variance Group B	Rp33.088.798.170,03
Variance Ratio	1,47

5. A/B Testing

Question

C. Is there any significant difference between Group A and Group B Transaction Value? (Hint – significant difference = hypothesis testing)

Answer

P(T<=t) two-tail < Level of Significance
0,00 < 0,05

Because the P-Value (0%) ≤ Alpha (5%)
→ Reject H_0 and the conclusion is H_1

There is **significant difference** between Group A and Group B Transaction Value

T Test: Two-Sample Assuming Equal Variances		
	Group A	Group B
Mean	Rp746.103	Rp830.460
Variance	22436486774	33088798170
Observations	498	502
Pooled Variance	27783989789	
Hypothesized Mean I	0	
df	998	
t Stat	-8,001901057	
P(T<=t) one-tail	1,69194E-15	
t Critical one-tail	1,646381816	
P(T<=t) two-tail	0,00	
t Critical two-tail	1,962343802	

5. A/B Testing

Question

D. Interpretation of Results & Business Implication

Answer

- Because the P-Value (0%) ≤ Alpha (5%) → Reject H_0 and the conclusion is H_1.

- This means that there is enough evidence to state that :
  transaction value per user new product detail page design ≠ transaction value per user current product detail page design

- Average **transaction value per user new** product detail page design **>** **current** product detail page.
- So we can release **new product detail page design** to all customer.

5. A/B Testing

Question

E. Provide a business recommendation on whether the new design should be rolled out to all users (Use OBIPR framework?)

Answer

Insights / Observation	Business Impact	Isolation	Prioritization	Recommendation
Average transaction value per user new product detail page design > current product detail page	Increase transaction value by 11%	Product detail page design	High (High impact – High effort – Many data point)	Release new product detail page design to all customer

Conclusion

- Campaign 12/12 yielded the best business results among other campaign. So we can apply Campaign 12/12 scenario for future campaign

- New product detail page design generate significantly higher transaction value per user than current product detail page. So we can release new product detail page design to all customer.

- Men’s Fashion was the best performing category. So we can push promotion on this category or bundling it with complementary category